

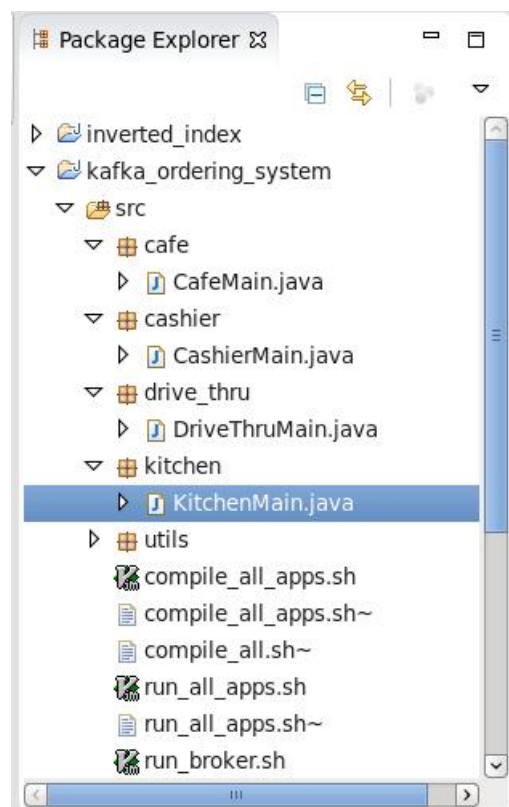
Lab Task 4

Write a MapReduce Program for Finding the Shifted Word Counts

COS20028 – Big data Architecture and Applications
Semester 2

Name: **Truong Ngoc Gia Hieu**
Student ID (Vietnam): **SWS01217**
Student ID (Australia): **105565520**
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Project Folder Structure Screenshot



Cashier.java Code

```
CafeMain.java  CashierMain.java  DriveThruMain.java  KitchenMain.java
1 package cashier;
2
3 import utils.OrderGenerator;
4
14 public class CashierMain {
15
16     public static void main(String[] args){
17         // Create a random order generator
18         Random random = new Random();
19         OrderGenerator dt_og = new OrderGenerator("C", C_SEED);
20
21         //-- Implementation. --//
22         Properties properties = new Properties();
23         properties.put(ProducerConfig.BOOTSTRAP_SERVERS_CONFIG, BOOTSTRAP_SERVER);
24         properties.put(ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG, StringSerializer.class.getName());
25         properties.put(ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG, StringSerializer.class.getName());
26         KafkaProducer<String, String> producer = new KafkaProducer<>(properties);
27
28         try{
29             for (int i = 0; i < NUM_ORDERS; i++){
30                 // Wait ...
31                 try{
32                     Thread.sleep((random.nextInt(6) + 10) * 1000);
33                 } catch (InterruptedException e){
34                     e.printStackTrace();
35                 }
36             }
37         }
```

Cafe.java Code

```
CafeMain.java  CashierMain.java  DriveThruMain.java  KitchenMain.java
1 package cafe;
2
3 import static utils.Constants.*;
4
5 import org.apache.kafka.clients.consumer.KafkaConsumer;
6 import org.apache.kafka.clients.consumer.ConsumerConfig;
7 import org.apache.kafka.clients.consumer.ConsumerRecord;
8 import org.apache.kafka.clients.consumer.ConsumerRecords;
9 import org.apache.kafka.common.serialization.StringDeserializer;
10
11 import java.time.Duration;
12 import java.util.Collections;
13 import java.util.Properties;
14 public class CafeMain {
15     public static void main(String[] args){
16         //-- IMplementation. --//
17         // Set properties and create a Kafka consumer
18         Properties properties = new Properties();
19         properties.put(ConsumerConfig.BOOTSTRAP_SERVERS_CONFIG, BOOTSTRAP_SERVER);
20         properties.put(ConsumerConfig.KEY_DESERIALIZER_CLASS_CONFIG, StringDeserializer.class.getName());
21         properties.put(ConsumerConfig.VALUE_DESERIALIZER_CLASS_CONFIG, StringDeserializer.class.getName());
22         properties.put(ConsumerConfig.GROUP_ID_CONFIG, FOOD_CONSUMER_GROUP);
23         properties.put(ConsumerConfig.AUTO_OFFSET_RESET_CONFIG, "earliest");
24         KafkaConsumer<String, String> consumer = new KafkaConsumer<String, String>(properties);
25
26         //Subscribe to the food topic
27     }
```

Screenshot of the Final Result Content

```
training@quickstart:~/workspace/kafka_ordering_system/src
File Edit View Search Terminal Help
[training@quickstart ~]$ pwd
/home/training
[training@quickstart ~]$ cd workspace
[training@quickstart workspace]$ ls
averagewordlength  kafka_game_score  mrunit          toolrunner
combiner           kafka_ordering_system  oozie-labs      wclibrary
counters          lab03              outputformat    word_co-occurrence
createsequencefile linecount          partitioner      wordcount
inputformat       log_file_analysis  secondarysort   word_co_writable
inverted_index    logging            test02          writables
[training@quickstart workspace]$ cd kafka_ordering_system
[training@quickstart kafka_ordering_system]$ cd src
[training@quickstart src]$ ls
cafe          compile_all_apps.sh~  kitchen        run_broker.sh
cashier       compile_all.sh~      run_all_apps.sh  utils
compile_all_apps.sh  drive_thru          run_all_apps.sh~
[training@quickstart src]$ sh compile_all_apps.sh~
[training@quickstart src]$ sh run_all_apps.sh
[training@quickstart src]$ sh run_all_apps.sh
[training@quickstart src]$

training@quickstart:~/workspace/kafka_ordering_system
File Edit View Search Terminal Help
p FOOD_GROUP in state PreparingRebalance with old generation 2 ( _consumer_offset=18) (reason: Adding new member consumer-FOOD_GROUP-1-0683fb1d-5862-4969-881d-b0ebb8d93cc0 with group instance id None; client reason: rebalance failed due to MemberIdRequiredException) (kafka.coordinator.group.GroupCoordinator)
[2026-07-13 20:19:56,556] INFO [GroupCoordinator 0]: Dynamic member with unknown member id joins group FOOD_GROUP in PreparingRebalance state. Created a new member id consumer-FOOD_GROUP-1-916fe4df-339b-46b9-b1e3-d71e7bd7b7af and request the member to rejoin with this id. (kafka.coordinator.group.GroupCoordinator)
[2026-07-13 20:20:21,830] INFO [GroupCoordinator 0]: Member consumer-FOOD_GROUP-1-369c1fcc-b28b-4262-b30a-5f99af658343 in group FOOD_GROUP has failed, removing it from the group (kafka.coordinator.group.GroupCoordinator)
[2026-07-13 20:20:21,836] INFO [GroupCoordinator 0]: Member consumer-FOOD_GROUP-1-18d76a35-7895-4308-9c4f-05b2101bbd0b in group FOOD_GROUP has failed, removing it from the group (kafka.coordinator.group.GroupCoordinator)
[2026-07-13 20:20:21,837] INFO [GroupCoordinator 0]: Stabilized group FOOD_GROUP

cafe
File Edit View Search Terminal Help
log4j:WARN No appenders could be found for logger (org.apache.kafka.clients.
umer.ConsumerConfig).
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for mor
fo.
C 1
[Matcha Tea Latte*1, Latte*1, Royal English British Tea Latte*1, Earl Grey T
atte*1]
DT 1
[Matcha Tea Latte*1, Latte*1, Cappuccino*1]
C 2
[Flat White*2, Earl Grey Tea Latte*1, Cappuccino*1]
DT 2
[Earl Grey Tea*1, Latte*1, Macchiato*1]
DT 3
[Flat White*1]
C 3
[Earl Grey Tea*1, Mocha*1]
C 4

kitchen
File Edit View Search Terminal Help
C 4
[Chocolate Brownie*1, Spinach & Feta Wrap*2]
DT 4
[Chicken Sandwich*1, Spinach & Feta Wrap*2]
DT 5
[Chicken Sandwich*1, Chocolate Brownie*2, Spinach & Feta Wrap*1]
C 5
[Bacon Sandwich*1, Spinach & Feta Wrap*1]
C 6
[Bacon Sandwich*1, Chocolate Brownie*2]
DT 7
[Chicken Sandwich*1, Spinach & Feta Wrap*3]
C 7
[Bacon Sandwich*1, Chocolate Brownie*1, Ham Croissant*1, Sausage & Egg Wrap*
utter Croissant*1]
```